



No difference in nitrate leaching between digestates and untreated cattle slurry over eight years of organic arable farming

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Abstract Organic fertilizers often contribute to nitrate (NO_3^-) leaching and impair drinking water quality, because the mineralization of applied organic nitrogen (N) is often asynchronous with crop N uptake. Crop yield, crop N uptake, seepage water and NO_3^- leaching were investigated in a lysimeter facility (Zurich, Switzerland) under temperate climate conditions during two consecutive four-year organically managed crop rotations. The study compared untreated cattle slurry, anaerobically digested slurry, liquid digestate from a non-agricultural biogas plant and a zero-N fertilization control. The ammonium N ($\text{NH}_4\text{-N}$) fraction of the organic fertilizers was labeled with ^{15}N tracer to investigate the fate of $\text{NH}_4\text{-N}$ in the plant-soil-water system. Organic fertilization significantly increased crop yields and N uptake, with no differences among organic fertilizers, despite the

higher $\text{NH}_4\text{-N}$ proportion of the digestates. Mean annual NO_3^- leaching was low ($12\text{--}18\text{ kg N ha}^{-1}$) and was not increased by organic fertilization, indicating that N inputs were balanced by crop N uptake. Digestates did not differ from untreated cattle slurry in NO_3^- leaching. Higher modeled ammonia (NH_3) emissions of digestates suggest that N losses may occur primarily via NH_3 volatilization rather than NO_3^- leaching. The ^{15}N recovery showed $\sim 1\%$ of the applied $\text{NH}_4\text{-N}$ was lost through NO_3^- leaching, whereas $23\text{--}31\%$ was recovered in crops, and $18\text{--}23\%$ remained in the topsoil. These results highlight the importance of crop N uptake and the role of the soil as a long-term reservoir for fertilizer-derived N, with direct NO_3^- leaching from the $\text{NH}_4\text{-N}$ fraction of organic fertilizers playing only a minor role.

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Introduction

Nitrate (NO_3^-) leaching into groundwater impairs drinking water quality (Di and Cameron 2002) and contributes to eutrophication of surface water bodies (Wassenaar 1995; Galloway et al. 2003; Zhang et al. 2015). Moreover, the presence of NO_3^- drives denitrification in groundwater (Jahangir et al. 2013) and surface waters, resulting in nitrous oxide (N_2O) emissions (Beaulieu et al. 2011). NO_3^- leaching from

agriculture is controlled, on the one hand, by anthropogenic factors, such as crop management, fertilizer type and application rate (Spiess et al. 2020; Bernert et al. 2025). On the other hand, NO_3^- leaching is affected by soil type, evapotranspiration and precipitation amount and distribution, which govern seepage water volumes (Di and Cameron 2002; Li et al. 2023). Under temperate climate conditions, nitrogen (N) in soil is prone to leaching in autumn and winter due to high seepage volume and low evapotranspiration along with small N uptake of crops (Di and Cameron 2002). Therefore, N applied or mineralized in autumn is at risk of being lost. In particular, NO_3^- leaching is a problem in intense farming systems with high excess N (Di and Cameron 2002; Velthof et al. 2024) and is directly linked to high livestock densities and excess manure production (Carpenter et al. 1998; Eriksen et al. 1999).

Organic farming aims to reduce the negative impact of intensive agriculture through the ban of synthetic mineral N fertilizers and improved nutrient cycling (Krause et al. 2024). However, organic farming systems tend to have lower yields compared to conventional farming systems (Seufert et al. 2012; Mayer et al. 2015), which may be mostly explained by the reduced availability of mineral N forms to organic crops (Knapp et al. 2023). Organic fertilizers as the sole fertilizers used in organic farming, show large differences in their N use efficiency (NUE) due to varying carbon-to-nitrogen ratio (C/N ratio) and plant-available ammonium N ($\text{NH}_4\text{-N}$) proportion (Gutser et al. 2005), resulting in reduced yields compared to mineral fertilization.

Manure as fertilizer input is expected to cause higher NO_3^- leaching, compared to mineral fertilizers due to asynchrony of mineralization of organically bound N and crop N uptake (Sørensen and Jensen 2013). While short-term NO_3^- leaching from manure can be minimal (Frick et al. 2022), repeated manure application increases the soil organic N pool compared to mineral fertilization only, which increases the long-term leaching risk (Bergström and Kirchmann 1999). Moreover, the use of manure and other organic resources is essential for closing nutrient cycles and replacing mineral N input (Velthof et al. 2024; Epper et al. 2025; Harder and Liebisch 2025), as well as increasing soil organic carbon (C) stocks (Krause et al. 2024).

Anaerobic digestion of manure is proposed to increase NUE and reduce N losses to the environment, due to its higher proportion of plant-available N compared to untreated liquid manure (Albuquerque, et al. 2012; Möller and Müller 2012). Anaerobic digestion of manure increases the proportion of $\text{NH}_4\text{-N}$ and decreases the organic C and organic N content due to the decomposition of organic compounds (Gutser et al. 2005; Möller and Müller 2012). Digestates are also characterized by higher pH values, due to the mineralization of organic N to NH_4^+ and the degeneration of fatty acids (Sommer and Husted 1995; Möller and Müller 2012), which increases the risk of ammonia (NH_3) emissions (Pedersen and Hafner 2023). The decomposition of organic matter also decreases the dry matter content in digestates, which has a positive effect on infiltration into the soil and therefore reduces the risk of NH_3 emissions. (Möller and Müller 2012). Anaerobic digestion further leads to reduced soil microbial $\text{NH}_4\text{-N}$ immobilization compared to untreated manures after field application, because of the relatively lower proportion of easily-degradable compounds (Jensen et al. 2000; Sørensen 2004). Overall, digestates tend to have a greater NUE in the year of application (Frick et al. 2024). However, varying effects have been reported due to the wide range of feedstock composition and co-substrate input, governing key characteristics (C/N-ratio, $\text{NH}_4\text{-N}$ proportion, dry matter content) of digestate for short-term N availability and NH_3 emissions (Möller and Müller 2012; Nkoa 2014). Likewise, anaerobic digestion also showed varying effects on NO_3^- leaching, with studies showing either similar (Zilio et al. 2023; Frick et al. 2024), reduced (Sogn et al. 2018) or even increased leaching compared to non-digested organic fertilizers (Jørgensen and Petersen 2006).

The objective of this study was to evaluate NO_3^- leaching from untreated cattle slurry, compared to an anaerobically digested slurry and a liquid digestate from a non-agricultural biogas plant digesting household and green waste. We hypothesized that the application of anaerobically digested liquid organic fertilizers in comparison to untreated slurry a) increases NUE and b) results in reduced NO_3^- leaching. To evaluate these hypotheses, we performed a lysimeter experiment in Zurich, Switzerland, over two consecutive four-year crop rotations between 2018 and 2025, in which the $\text{NH}_4\text{-N}$ fraction of the

fertilizers was labeled with a ^{15}N tracer. In addition, a mirrored field experiment with the same treatment design was used to investigate similar treatment differences in N translocation along a soil profile.

Materials & methods

Lysimeter facility

The experiment was conducted on Agroscope’s lysimeter facility based at Zurich (Switzerland, 47.428056, 8.518056, 440 m a.s.l.) from 2018 to 2025. The lysimeters contained soil monoliths of a Cambisol from Grafenried (Switzerland, canton Berne) collected in 2008, with a surface of 1 m² and an effective soil depth of 1.35 m. The base of the lysimeter was filled with 0.15 m of quartz sand in three layers of different particle sizes (0.10–0.50 mm, 0.71–1.25 mm, and 3.15–5.60 mm) to prevent water logging, which would favor denitrifying conditions. Soil texture in the top 0.2 m was 16% clay, 32% silt and 51% sand, with initial concentrations of soil organic C of 17 g kg^{−1} and soil organic N of 1.1 g kg^{−1} (Spiess et al. 2015). Annual mean temperature and precipitation in the 30-year reference period (1988–2017) were 9.8 °C and 1029 mm, respectively. Before the start of the experiment in 2018, the lysimeters used in this study followed the crop rotation winter wheat-triticale and were minerally fertilized with NH₄NO₃ according to Swiss fertilization standards (PRIF, Sinaj and Richner, 2017) without the addition of organic fertilizers.

The experiment spanned two consecutive four-year crop rotations including maize (*Z. mays* L.), winter wheat (*Triticum aestivum* L.), winter barley (*Hordeum*

vulgare L.), and forage grass mixture (*Lolium perenne* L., *Lolium multiflorum* Lam., *Lolium* × *hybridum* Hausskn., *Festuca pratensis* Huds.). After the harvest of winter wheat in early summer, yellow mustard was immediately sown as cover crop and mulched before seedbed preparation for winter barley (Supplemental Information 1). The experiment followed the rules of organic farming standards (Bio Suisse 2023), and no plant protection products or growth regulators were applied. Soil and crop management (weeding, plowing) on the lysimeters was done by hand (Supplemental Information 1).

Weather conditions during the experiment

Weather data were monitored by a weather station located on site and operated by MeteoSwiss (47.427653, 8.517886). Average, minimal, and maximal temperatures (2 m above ground) and precipitation were measured hourly and aggregated to daily values.

Table 1 shows the seasonal precipitation during the experiment with the 30-year average as reference. Precipitation is highest during summer months, and lowest during winter and autumn, which is characteristic of the local temperate climate. The years 2018, 2020 and 2022 were characterized as dry years, because the annual precipitation deficit exceeded one standard deviation (152 mm) of the long-term mean (Table 1). By contrast, 2024 was a wet year with the precipitation surplus exceeding the standard deviation of the long-term mean. The years 2019, 2021, 2023 and 2025 can be considered as normal, since the precipitation was within natural variation (± 1 standard deviation of the long-term mean). Generally, in most of the investigated years (2018, 2019, 2022,

Table 1 Seasonal average precipitation at Zurich with the reference of the 30-year mean (1988–2017). Winter includes December of the previous year, January and February; spring includes March, April, and May; summer includes June, July,

and August; and autumn includes September, October, and November. Bold values indicate values above the 30-year average

Precipitation (mm)	1988—2017	2018	2019	2020	2021	2022	2023	2024	2025
Winter	240	170	273	203	270	196	150	294	242
Spring	261	153	252	158	238	159	274	352	198
Summer	335	262	308	320	434	230	318	296	331
Autumn	193	146	248	153	71	280	294	314	294
Year	1029	731	1081	834	1013	865	1036	1256	1065

2023, 2024, 2025), summers were drier than average, whereas autumns were wet (2019, 2022, 2023, 2024, 2025).

Fertilizer characteristics and application

Untreated cattle slurry (SLU), anaerobically digested slurry (SLA), liquid digestate from a non-agricultural biogas plant (LID), and a zero-N fertilization control (NON) were investigated in a full randomized design with three replicates. The fertilizers applied on the lysimeters were also applied in a mirrored field trial in Wallbach (47.562306, 7.887694, Efosa et al. 2023). Fertilized treatments received the same amount of total N following Swiss fertilization recommendations (Sinaj and Richner 2017). Maize, winter wheat and winter barley were fertilized in two equal split applications. Total N application rates were 140 kg N ha⁻¹ to both maize and winter wheat and 120 kg N ha⁻¹ to winter barley. Forage grass was fertilized with 50 kg N ha⁻¹ per application, resulting in total N application rates of 300 kg N ha⁻¹ in 2021 (six applications) and 200 kg N ha⁻¹ in 2024 (four applications), due to different cutting frequencies. NON received P (20 kg ha⁻¹) and K (130 kg ha⁻¹) fertilization annually, corresponding to similar P and K inputs as the organic fertilizers. LID had a lower K content than SLA and SLU and was annually supplied with 65 kg ha⁻¹ K as mineral fertilizer (Supplemental Information 3). Organic fertilizers were applied manually using a measuring cylinder in rows spaced approximately 0.25 m apart, and from a height of 0.1 m above the soil surface to mimic application with a trailing hose.

SLU was sourced from organic farms in North-western Switzerland, and consisted of excreta, straw used as bedding and rainwater. The origin of SLU was different in 2024 and 2025, because of the installation of a separator at the initial farm. SLA originated

from an agricultural biogas plant, that digested cattle slurry and horse manure and various food-processing residues (e.g., cacao chips from chocolate production) as co-substrates. Since the NH₄-N proportion of SLA from this plant did not differ greatly from SLU, SLA was sourced from another biogas plant in 2024 and 2025, which used cattle slurry and vegetable leftovers as co-substrates. LID was obtained from a commercial non-agricultural biogas plant that digested plant material from urban green areas and organic waste from households, supermarkets, food processing industries, and gastronomy. The organic wastes were subjected to thermophilic anaerobic digestion at 55 °C, with a mean residence time of 15 days. SLA and LID were the liquid fractions resulting from the solid–liquid separation of the respective digestate using a screw press.

Fertilizer aliquots were cooled directly after application and analyzed at ‘Labor für Boden- und Umweltanalytik’ (Thun, Switzerland). Table 2 shows the mean values of the fertilizer characteristics of the treatments, whereas yearly values are found in Supplemental Information 2.

¹⁵N labeling of fertilizers applied at the lysimeter facility

The NH₄-N fraction of the organic fertilizers (SLU, SLA and LID) was labeled directly before application to the lysimeter with ammonium sulphate (¹⁵NH₄)₂SO₄ tracer solution (99 atom%) to an enrichment of 7 atom% ¹⁵N excess in three different years (2018, 2019, 2024). For each lysimeter, the volume needed to obtain the target amount of N per application (e.g., 7 g N m⁻¹ equivalent to 70 kg N ha⁻¹ per application to winter wheat) was taken and mixed thoroughly with the corresponding amount of (¹⁵NH₄)₂SO₄ solution. A subsample was collected for

Table 2 Total nitrogen concentration (N_{tot}, g kg⁻¹ FW), NH₄-N concentration (g kg⁻¹ FW), NH₄-N proportion (% of N_{tot}), dry matter (DM, %), pH value and C/N ratio of untreated slurry (SLU), anaerobically digested slurry (SLA), and liq-

uid digestate from non-agricultural biogas plant (LID). Values show averages of all applied organic fertilizers during the experiment with standard deviation (n = 22)

Treatment	N _{tot} (g kg ⁻¹ FW)	NH ₄ -N (g kg ⁻¹ FW)	NH ₄ -N (%)	Dry matter (%)	pH	C/N ratio
SLU	1.7 ± 0.4	0.8 ± 0.2	49.6 ± 5.1	4.2 ± 1.6	7.4 ± 0.3	9.6 ± 2.0
SLA	3.7 ± 0.9	2.2 ± 0.6	60.3 ± 7.0	5.1 ± 1.3	8.0 ± 0.3	4.9 ± 1.0
LID	5.5 ± 0.4	3.1 ± 0.2	55.9 ± 2.3	10.4 ± 2.2	7.9 ± 0.1	6.5 ± 1.0

analysis, before applying the required volume to the lysimeter.

Plant sampling

The entire surface area (1 m²) of the lysimeter was harvested to quantify the yields and N uptake of the crops. The silage maize was harvested at the dough stage (BBCH 85), whereas winter wheat and winter barley were harvested at full maturity (BBCH 89). Forage grass was harvested at heading (BBCH 55).

The fresh weight of all samples was assessed before maize was separated into cob and shoot, and cereals into ears and shoot. The ears were threshed to obtain the grain yield. All samples were then dried at 60 °C for 72 h to obtain the dry matter yield. The samples were first milled on a cutting mill (Retsch SM300, Germany) before further homogenization with a ball mill (Retsch MM200, Germany).

Seepage sampling and NO₃[−] measurement

The seepage water volume was recorded automatically with tipping buckets. Since the surface area of the lysimeters was 1 m², 1 mm of seepage water was equal to 1 l m^{−2}. Seepage water was analyzed photometrically biweekly for NO₃-N concentration (mg l^{−1}) by segment-flow injection analysis. Forage grass establishment was unsuccessful in 2020 on two lysimeters (Supplemental Information 8), and it had to be resown; therefore, the data were replaced by the mean values of the other replicates of the treatments during winter 2020/2021. NO₃-N leaching was calculated per unit of area (kg N ha^{−1}) with Eq. 1:

$$\text{NO}_3 - \text{N}_{\text{leached}} (\text{kg N ha}^{-1}) = \text{Seepage volume (l m}^{-2}) \times \text{NO}_3\text{N (mg l}^{-1}) \div 100 \quad (1)$$

Soil sampling in the lysimeter

At the end of the experimental period (December 2025), samples were obtained from 0.2 m topsoil of the lysimeters. From every lysimeter, 10 samples were taken using a small soil sampler (diameter 10 mm) and mixed into one composite sample. Soil samples were sieved freshly with a 2 mm mesh, and small roots and fresh plant material was removed manually before drying at 60 °C for 72 h and homogenization with a ball mill (Retsch MM200, Germany). Since,

destructive soil sampling was not possible, bulk soil density was assumed to be 1.4 g cm^{−3}, regardless of the treatments.

Soil sampling of ¹⁵N labeled microplots in the field

The lysimeter experiment mirrored a field trial in Wallbach (47.56225° N, 7.88766° E, Efosa et al. 2023) with the same treatments. The field trial followed a split-plot block design with four replicates over the same crop rotation including maize (*Z. mays* L.), winter wheat (*Triticum aestivum* L.), winter barley (*Hordeum vulgare* L.), and forage grass mixture (*Lolium perenne* L., *Lolium multiflorum* Lam., *Lolium* × *hybridum* Hausskn., *Festuca pratensis* Huds.). The same batches of organic fertilizers as those applied in the lysimeters (Supplemental Information 2) were applied in the field, typically within two days prior to the application in the lysimeters.

In the field, 3 × 2.25 m microplots within each fertilizer plot were designated for ¹⁵N labeling. Isotopic labeling was performed in 2018, 2019 and 2024 in different microplots, to investigate the residual effect of ¹⁵N labelled fertilizers in subsequent years. The microplot labeled in 2018 is referred to as MP18, whereas the microplot labeled in 2019 and 2024 is referred to as MP19 and MP24, respectively. Due to a change in the research focus in the field, LID in MP24 was not labeled with ¹⁵N. The NH₄-N fraction of the fertilizer was labeled with (¹⁵NH₄)₂SO₄ (99 atom%) tracer solution to an enrichment of 7 atom% ¹⁵N excess. The volume of the target N application for each individual plot was filled into a barrel, and the pre-calculated tracer solution was added before stir-

ring firmly with a wooden stick. A subsample was taken for later analysis before applying the entire volume with a watering can 0.1 m above the soil surface to mimic a trailing hose.

A minimum of three soil cores to a depth of 1.0 m were taken with a “Pürckhauer” auger within the MPs in February 2025, when forage grass was grown. The cores were separated manually into four soil layers (0 – 0.2 m, 0.2 – 0.4 m, 0.4 – 0.6 m and 0.6 – 1.0 m) and combined into a composite sample per layer. Samples were cooled until sieving

with a 2 mm mesh. A fresh subsample equivalent to ~ 100 g fresh weight was dried at 60 °C for 72 h. After achieving constant weight, a subsample was milled on a ball mil (Retsch MM200, Germany) to fully homogenize the sample.

¹⁵N samples preparation in the lab and analysis

The ¹⁵N diffusion technique (Goerges and Dittert 1998) was applied to the seepage water samples and fertilizer samples to determine ¹⁵N enrichment. NO₃⁻ and NH₄⁺ in seepage water samples were diffused onto acidified filter traps (Whatman

filters from labeled fertilizers was measured on an elemental analyzer (Euro EA, EuroVector, Italy) coupled with an isotope ratio mass spectrometer (Delta XP, Thermo Electron, Germany) at the Center for Stable Isotope Research and Analysis at the University of Göttingen (Germany).

Nitrogen use efficiency and ¹⁵N calculations

Apparent NUE (%) was calculated using the difference method (Eq. 2) as the difference between above-ground N uptake of fertilized crops and N uptake of unfertilized crops in comparison to N applied, with all values expressed in kg N ha⁻¹ (Jensen 2013; Epper et al. 2025).

$$\text{Apparent NUE (\%)} = \frac{\text{N uptake}_{\text{fertilized}} (\text{kg ha}^{-1}) - \text{N uptake}_{\text{NON}} (\text{kg ha}^{-1})}{\text{N}_{\text{applied}} (\text{kg ha}^{-1})} \quad (2)$$

QM/A), which were separated by a Teflon tape from the sample solution inside a Nalgene bottle. Diffusion was performed by adding Devarda's alloy (0.45 g per sample, reducing agent to convert NO₃⁻ to NH₄⁺), 4 mL of 5 M NaCl, and 0.75 mL of 5 M NaOH per 10 mL of seepage water sample (Goerges and Dittert 1998; Mayer et al. 2003) and shaking for 48 h on a horizontal shaker.

Labeled fertilizer samples were initially centrifuged at 4600 rpm for 15 min to separate the solid and liquid fractions. The remaining liquid fractions were then diffused on acidified filter traps by adding 4 mL of 5 M NaCl, and 0.75 mL of 5 M NaOH per 10 mL of sample (Goerges and Dittert 1998; Mayer et al. 2003). Devarda's alloy was not added to prevent the reduction of NO₃⁻ to NH₄⁺, ensuring that only the NH₄-N fraction of the fertilizers was captured by diffusion.

The N concentration and ¹⁵N enrichment of the lysimeter soil and plant samples, the microplot soil samples, and the diffusion filters of seepage water samples were analyzed on an elemental analyzer (PYRO cube, Elementar, Germany) coupled with an isotope ratio mass spectrometer (isoprime precision, Elementar, Germany) at Agroscope (Switzerland). The high ¹⁵N enrichment of the diffusion

For all ¹⁵N data, the year of ¹⁵N labeling was defined as year zero (Yr₀), the first year after the ¹⁵N application as residual year 1 (R₁), and so on (Macdonald et al. 2002; Smith and Chalk 2018).

Atom ¹⁵N excess was calculated by subtracting the ¹⁵N abundance of the non-labeled reference material of the corresponding compartment (plant, soil, seepage water) from the measured ¹⁵N abundance of each sample. The percentage of N derived from fertilizer-ammonium N (Ndffa) in the corresponding compartment (Eq. 3) was calculated according to Hauck and Bremner (1976):

$$\text{Ndffa (\%)} = \frac{\text{atom\% } ^{15}\text{N}_{\text{excess sample}}}{\text{atom\% } ^{15}\text{N}_{\text{excess NH}_4\text{N - fertilizer}}} \times 100 \quad (3)$$

Since only the NH₄-N fraction of the fertilizer was labeled, only the percentage of Ndffa could be assessed. The absolute amount of Ndffa in kg N ha⁻¹ in the compartment was calculated using Eq. 4, where N total_{comp} is the total N amount in the compartment considered (Hood et al. 2000).

$$\text{Ndffa (kg ha}^{-1}\text{)} = \frac{\text{Ndffa(\%)}}{100} \times \text{N total}_{\text{comp}} (\text{kg ha}^{-1}) \quad (4)$$

Finally, NH₄-N-fertilizer recovery of the respective compartment was calculated using Eq. 5 as:

$$\text{NH}_4 - \text{N} - \text{recovery} (\%) = \frac{\text{Ndffa} (\text{kg ha}^{-1})}{\text{NH}_4 - \text{N applied} (\text{kg ha}^{-1})} \quad (5)$$

Modeled NH_3 emissions at application

NH_3 emissions at application were modeled with the ALFAM2 model (version 4.2.14) using parameter set 3 (Hafner et al. 2025). Proportions of $\text{NH}_4\text{-N}$, fertilizer application rate, fertilizer dry matter and fertilizer pH, were provided to the model for each application individually. Temperature (2 m above ground), precipitation and windspeed (10 m above ground) were used with a temporal resolution of 1 h from a nearby weather station (MeteoSwiss, Switzerland). Windspeed was transformed from 10 to 2 m following the logarithmic wind profile described by Allen et al. (1998). Cumulative NH_3 emissions were modeled for 72 h (Efosa et al. 2025).

$\text{NH}_4\text{-N}$ balance

A $\text{NH}_4\text{-N}$ balance over the experimental period was calculated using Eq. 6, based on ^{15}N labeled fertilizer input and output:

$$\begin{aligned} \text{NH}_4 - \text{N} - \text{Balance} (\text{kg ha}^{-1}) = & \text{NH}_4\text{N applied} (\text{kg ha}^{-1}) \\ & - \text{NH}_3 \text{ emissions} (\text{kg ha}^{-1}) - \text{Ndffa}_{\text{crop}} (\text{kg ha}^{-1}) \\ & - \text{Ndffa}_{\text{seepage}} (\text{kg ha}^{-1}) - \text{Ndffa}_{\text{soil}} (\text{kg ha}^{-1}) \end{aligned} \quad (6)$$

where $\text{NH}_4\text{N applied}$ is the amount of $\text{NH}_4\text{-N}$ applied (kg N ha^{-1}) in the years of ^{15}N labeling (2018, 2019, 2024) and NH_3 emissions are the modeled NH_3 losses at application in these years. We assumed, that all NH_3 is a direct loss of $\text{NH}_4\text{-N}$ and therefore also of the ^{15}N -labeled pool. Ndffa (kg ha^{-1}) in crops ($\text{Ndffa}_{\text{crop}}$) and in seepage water ($\text{Ndffa}_{\text{seepage}}$) was the sum of all years (2018–2025), and $\text{Ndffa}_{\text{soil}}$ was the final ^{15}N recovered in the soil at the end of the experiment (2025).

Statistical analysis

All statistical analyses were conducted using R (R Core Team 2021, version 4.4.1), with the use of the packages *dplyr* (Wickham et al. 2026) for data handling and *ggplot2* (Wickham 2016) for plotting. A

linear mixed effects model using the *lmer* function from the R-package *lme4* (Bates et al. 2015) was used to analyze the fixed effects of treatments on target variables (dry matter, N uptake, apparent NUE, seepage water, NO_3^- concentration, NO_3^- leaching). The individual lysimeter (plot) and year were set as random effects to account for the repeated measures within the same plot. The model fit was assessed using the marginal coefficient of determination ($R^2\text{m}$) and conditional R^2 ($R^2\text{c}$), explaining variance by fixed effects alone and fixed and random effects, respectively. The significance of the random effects was tested with an analysis of variance (ANOVA) type III. Estimated marginal means (*emmeans*) adjusted by Tukey's honestly significant difference (Tukey's HSD) were used to make pairwise comparisons across all years, accounting for random effects. Within years, analysis of variance was performed followed by Tukey's HSD post-hoc test. The significance level was set to $\alpha = 0.05$ throughout all analyses.

Results

Yields, N uptake and nitrogen use efficiency of crops

The organic fertilizers (SLU, SLA, LID) significantly increased dry matter yield (t ha^{-1}) and N uptake (kg N ha^{-1}) compared to NON across the entire experimental period ($p < 0.05$, Tukey's HSD, Table 3). Marginal R^2 , explaining the variance of fixed effects was low for dry matter yield ($R^2\text{m} = 0.29$, $R^2\text{c} = 0.86$) and N uptake ($R^2\text{m} = 0.22$, $R^2\text{c} = 0.84$). By contrast, conditional R^2 , explaining the variance of random effects, was high, indicating that the year and plot explained most of the variance. Although LID showed the lowest cumulative overall dry matter yield and N uptake of the fertilizer treatments (Table 3), no significant differences among fertilizer treatments were observed ($p > 0.05$, TukeyHSD). Across crop rotations, dry matter yield and N uptake of winter wheat and forage grass were significantly higher ($p < 0.05$, Tukey's HSD) with organic fertilization, whereas for winter barley only SLA differed significantly from NON (Fig. 1). Maize showed substantial variation in dry matter yield and N uptake between the two crop rotations, resulting in no significant differences between NON and the fertilizer treatments ($p > 0.05$, Tukey's HSD).

Table 3 Cumulative sum of applied total N (kg N ha^{-1}), dry matter yield (t ha^{-1}), N uptake (kg N ha^{-1}), apparent nitrogen use efficiency (NUE, %), seepage water (mm) and leached NO_3^- (kg N ha^{-1}) from lysimeters under different fertilization treatments: zero-N fertilization control (NON), untreated

	N applied (kg N ha^{-1})	Dry matter yield (t ha^{-1})	N uptake (kg N ha^{-1})	Apparent NUE (%)	Seepage water (mm)	NO_3^- leached (kg N ha^{-1})
NON	0	51.2 ± 1.9 (b)	437.1 ± 17.0 (b)	NA	3831 ± 40 (a)	127.4 ± 18.2 (ab)
SLU	1300	89.7 ± 4.0 (a)	806.4 ± 47.9 (a)	25.4 ± 15.4 (a)	3084 ± 24 (b)	133.9 ± 18.5 (ab)
SLA	1323	93.9 ± 3.2 (a)	803.9 ± 33.1 (a)	28.0 ± 12.7 (a)	3202 ± 109 (b)	145.6 ± 25.5 (a)
LID	1347	85.6 ± 3.2 (a)	729.6 ± 35.0 (a)	20.7 ± 7.4 (a)	3094 ± 156 (b)	96.4 ± 8.5 (b)

slurry (SLU), anaerobically digested slurry (SLA), and liquid digestate from non-agricultural biogas plant (LID) during the experimental period. Values show mean values with standard deviation

There were no significant differences in apparent NUE between the fertilizer treatments ($p > 0.05$, ANOVA type III). Marginal R^2 for apparent NUE, explaining the variance of fixed effects (treatments), was low ($R^2_m = 0.05$, $R^2_c = 0.61$). The forage grass in 2025 was the only single year with significantly higher apparent NUE for SLU compared to LID (Supplemental Information 5, $p > 0.05$, Tukey's HSD).

Seepage water and NO_3^- leaching over eight years

During the experimental period from 2018 to 2025, the highest total seepage water was recorded under NON, with a cumulative amount of 3831 mm, followed by SLA (3202 mm), LID (3094 mm), and SLU (3084 mm) (Table 3). Seepage water was significantly higher under NON compared to the fertilizer treatments (Tukey's HSD, $p < 0.05$), whereas no significant difference between the organic fertilizer treatments was observed. The standard deviation of seepage water was high for LID and SLA, indicating remarkable differences between individual lysimeters. The average value of seepage water over the eight years was 479 mm yr^{-1} in NON (Supplemental Information 6), which equals 46% of the annual precipitation (1029 mm yr^{-1}). In the fertilizer treatments annual seepage water was significantly lower ranging from 386 mm yr^{-1} (38% of annual precipitation) under SLU to 400 mm yr^{-1} (39% of annual precipitation) under SLA.

Over the eight-year experimental period the highest NO_3^- leaching occurred under SLA with

$145.6 \text{ kg N ha}^{-1}$, followed by SLU with $133.9 \text{ kg N ha}^{-1}$ and NON with $127.4 \text{ kg N ha}^{-1}$ (Table 3). The lowest NO_3^- leaching was observed with LID, totaling $96.4 \text{ kg N ha}^{-1}$, being significantly lower than for SLA. The annual mean NO_3^- leaching was maximally $18 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ for SLA and minimally $12 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ for LID (Supplemental Information 7). However, NON showed similar mean annual NO_3^- leaching of $16 \text{ kg N ha}^{-1} \text{ yr}^{-1}$. The mixed-effect model had a low marginal R^2 ($R^2_m = 0.035$) but high conditional R^2 ($R^2_c = 0.803$) indicating that fixed effects of treatments explained only a small proportion in annual NO_3^- leaching, whereas most of the explained variation was attributed to random effects (year and lysimeter plot).

Seasonal pattern of seepage water and NO_3^- leaching

Seepage water (mm), NO_3^- -N concentration (mg L^{-1}) and NO_3^- -N leached (kg N ha^{-1}) followed a clear seasonal pattern, with high mean values during winter and small amounts during summer (Table 4). The standard deviation for seepage water was high especially in summer, indicating large variation between years. In all seasons, NON led to significantly higher seepage water ($p < 0.05$, Tukey-adjusted) than the liquid organic fertilizers. No seasonal differences in seepage water were observed among the organic fertilizer treatments ($p > 0.05$, Tukey-adjusted).

Treatment differences in NO_3^- -N concentration were affected by the season, with the highest NO_3^- -N concentration in summer for SLA, and in autumn for

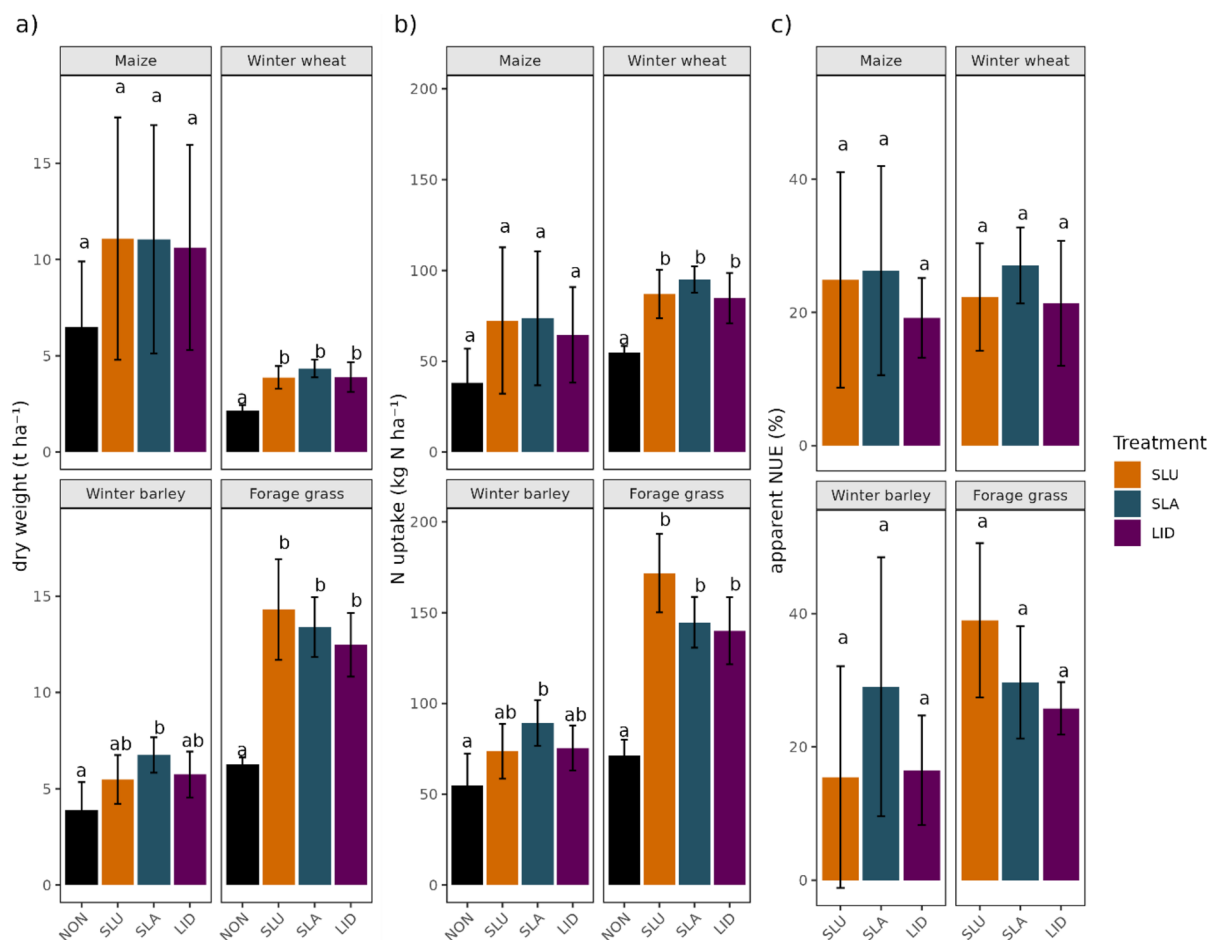


Fig. 1 Mean values ($\pm$ SD) of a) harvested dry matter yield (t ha^{-1}); b) N uptake (kg N ha^{-1}) across two crop rotations grouped by same crop from lysimeters under different fertilization treatments: zero-N fertilization control (NON), untreated slurry (SLU), anaerobically digested slurry (SLA), and liquid digestate from non-agricultural biogas plant (LID); and c) apparent nitrogen use efficiency (NUE, %) of SLU, SLA, LID.

Dry matter yield of winter wheat and winter barley includes only grains, whereas maize is the entire plant and forage grass is the sum of all cuts. N uptake includes the total N uptake of aboveground biomass. Different letters indicate significant differences between treatments ($p > 0.05$, Tukey-adjusted) based on estimated marginal means from a linear mixed-effect model with year and plot as random effect

SLU ($p < 0.05$, Tukey-adjusted). During winter, SLA and SLU had higher $\text{NO}_3\text{-N}$ concentrations compared to LID, but not compared to NON ($p < 0.05$, Tukey-adjusted).

The amounts of leached $\text{NO}_3\text{-N}$ showed treatment differences in winter and spring. In winter, significantly more $\text{NO}_3\text{-N}$ was leached under SLU and SLA than under LID, but no difference to NON was observed. In spring, more $\text{NO}_3\text{-N}$ was leached under NON than under LID ($p < 0.05$, Tukey-adjusted).

The cumulative sum of seepage water differed between NON and the fertilizer treatments as early

as the first winter of the experiment (2018/2019). In the course of the experimental period, more water was drained under NON every winter, and the difference compared to the fertilized treatments increased. From 2018 until 2023, the differences in NO_3^- leaching between treatments mirrored the seepage water patterns and followed the order $\text{LID} < \text{SLU} < \text{SLA} < \text{NON}$. However, the pattern changed from 2023 onwards, as the amount of leached NO_3^- increased proportionally more in the fertilized treatments than in NON. From summer 2021 until autumn 2023 NO_3^- leaching was very low ($< 2 \text{ kg N ha}^{-1} \text{ yr}^{-1}$),

Table 4 Cumulative sum of seepage water (mm), NO_3^- concentration (mg L^{-1}) and cumulative leached NO_3^- (kg N ha^{-1}) per season from lysimeters under different fertilization treatments: zero-N fertilization control (NON), untreated slurry (SLU), anaerobically digested slurry (SLA), and liquid digestate from non-agricultural biogas plant (LID) during the experimental period. Spring includes the months March, April,

and May; summer includes June, July, and August; autumn includes September, October, and November; winter includes December, January, and February. Numbers show mean values with standard deviation. Different letters indicate significant differences between treatments ($p > 0.05$, Tukey-adjusted) based on estimated marginal means from a linear mixed-effect model with year as random effect

Season	Treatment	Seepage water (mm)	$\text{NO}_3\text{-N}$ concentration (mg L^{-1})	$\text{NO}_3\text{-N}$ leached (kg N ha^{-1})
Spring	NON	78 ± 41 a	2.46 ± 3.52 a	2.2 ± 2.8 a
	SLU	64 ± 39 b	2.41 ± 2.63 a	1.8 ± 2.0 ab
	SLA	63 ± 37 b	2.58 ± 3.05 a	1.8 ± 1.8 ab
	LID	64 ± 40 b	1.46 ± 1.75 b	1.2 ± 1.4 b
Summer	NON	68 ± 93 a	1.15 ± 1.50 b	0.6 ± 0.6 a
	SLU	41 ± 60 b	1.08 ± 1.47 b	0.4 ± 0.6 a
	SLA	44 ± 69 b	2.64 ± 2.98 a	0.7 ± 1.2 a
	LID	42 ± 66 b	1.00 ± 2.03 b	0.2 ± 0.2 a
Autumn	NON	89 ± 79 a	3.00 ± 2.70 b	3.5 ± 3.8 a
	SLU	61 ± 62 b	5.20 ± 3.99 a	3.1 ± 4.1 a
	SLA	65 ± 62 b	3.74 ± 2.70 b	3.6 ± 4.3 a
	LID	60 ± 64 b	3.30 ± 2.57 b	2.2 ± 3.0 a
Winter	NON	243 ± 44 a	3.83 ± 3.83 ab	9.5 ± 7.4 ab
	SLU	220 ± 24 b	4.78 ± 4.38 a	11.5 ± 8.1 a
	SLA	227 ± 33 b	4.71 ± 4.38 a	12.0 ± 8.6 a
	LID	220 ± 31 b	3.48 ± 3.19 b	8.5 ± 6.3 b

due to $\text{NO}_3\text{-N}$ concentrations close to zero in the seepage water (Fig. 2).

^{15}N derived from fertilizer in crops

The ^{15}N derived from fertilizer-ammonium N (Ndffa) in crops and $\text{NH}_4\text{-N}$ recovery was highest in the year of ^{15}N tracer application (2018, 2019, and 2024), followed by a strong decrease in the residual years (Fig. 3). Ndffa in the first year of application ranged between only 4.5 kg N ha^{-1} (SLU in 2018) and 31.3 kg N ha^{-1} (SLA in 2019), due to varying productivity between years. Analogously, $\text{NH}_4\text{-N}$ recovery in the first year of application was between 6.8% (SLU 2018) and 38.3% (SLA in 2024). In all the years with ^{15}N application, $\text{NH}_4\text{-N}$ recovery followed the order $\text{SLU} < \text{LID} < \text{SLA}$, with significant differences between SLU and SLA in the years 2018 and 2019.

In the first residual year after ^{15}N application (2020), $\text{NH}_4\text{-N}$ recovery was less than 3%, and Ndffa was about 2 kg N ha^{-1} for all treatments. In

the second residual year (2021), $\text{NH}_4\text{-N}$ recovery did not further decrease, and was in the same order as in 2020 due to the high N uptake of the forage grass and accounted for about 3%. In subsequent years the residual Ndffa further decreased to ~1 kg N ha^{-1} in 2023. In 2025, the first residual year after ^{15}N tracer application in 2024, the $\text{NH}_4\text{-N}$ recovery of forage was 5–7%. $\text{NH}_4\text{-N}$ recovery in the residual years did not differ among treatments, except in 2021 and 2025, when forage grass was cultivated and recovery was significantly higher in SLU.

^{15}N derived from fertilizer in seepage water

Leached ^{15}N recovered in seepage water over eight years was low ($\text{SLA} = 3.0 \pm 0.6 \text{ kg N ha}^{-1}$, $\text{SLU} = 2.4 \pm 0.3 \text{ kg N ha}^{-1}$, $\text{LID} = 2.3 \pm 0.8 \text{ kg N ha}^{-1}$) and no significant differences between fertilizer treatments were observed ($p > 0.05$). ^{15}N leaching under LID was lowest, but the difference between fertilizers was not statistically significant. The year affected recovered ^{15}N in seepage water significantly (Tukey

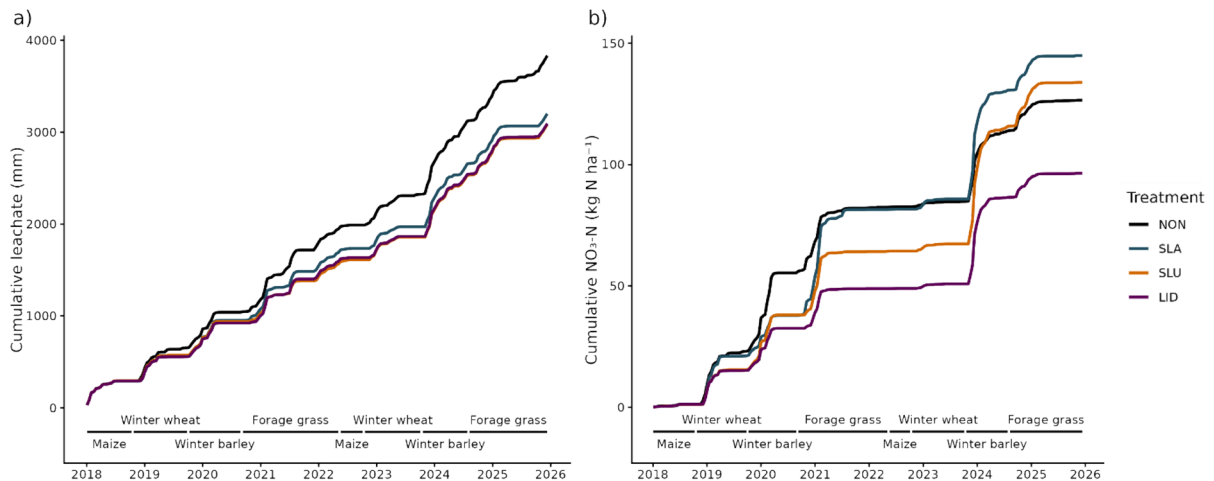


Fig. 2 Cumulative sum of **a** seepage water and **b** leached NO_3^- between the start of 2018 and the end of 2025 from lysimeters under different fertilization treatments: zero-N fertilization control (NON), untreated slurry (SLU), anaerobically

digested slurry (SLA), and liquid digestate from non-agricultural biogas plant (LID). Lines indicate the mean values of the treatments. The crop cultivated in each period is represented at the bottom

HSD, $p < 0.05$), with the highest but small recoveries ($< 1 \text{ kg N ha}^{-1}$) in the years of ^{15}N labeling (2019, 2024, Fig. 4). In the first year of labeling (2018), no immediate leaching was observed, but in the following year. Recovered ^{15}N in seepage water followed a clear seasonal trend: After ^{15}N labelling, little ^{15}N was directly leached after application (e.g., March 2024), followed by almost no ^{15}N leaching throughout the summer, although seepage water was present, and increasing ^{15}N leaching during winter.

^{15}N recovered in lysimeter soil

After eight years of experiment, soil organic C and N concentrations in the topsoil (0.0–0.2 m) of the lysimeter were higher for the organically fertilized treatments compared to NON, indicating a decrease in soil organic C and N concentrations without the use of organic fertilizers (Table 5). A considerable amount of the ^{15}N labeled $\text{NH}_4\text{-N}$ was recovered in the topsoil (0.0–0.2 m), ranging from $44.9 \text{ kg N ha}^{-1}$ in SLU to $48.58 \text{ kg N ha}^{-1}$ in LID. Accordingly, 18.2–23.2% of the applied $\text{NH}_4\text{-N}$ was recovered in the topsoil after eight years (Table 5).

$\text{NH}_4\text{-N}$ balance over eight years

The $\text{NH}_4\text{-N}$ balance revealed that, for all treatments, the losses through NO_3^- leaching were minimal with values of 2–3 kg N ha^{-1} (1% of applied $\text{NH}_4\text{-N}$, Fig. 5). Recovery in the soil was about 20% for all treatments (SLU: 23.2%, SLA: 18.2%, LID: 21.5%), whereas NH_3 losses and crop uptake varied between fertilizer treatments. The biggest losses through NH_3 volatilization at application were observed in LID ($61.9 \text{ kg N ha}^{-1}$, 27.4%), whereas the recovery of $\text{NH}_4\text{-N}$ in soil was highest in SLU ($44.9 \text{ kg N ha}^{-1}$, 23.2%) and the recovery of $\text{NH}_4\text{-N}$ in crop was highest in SLA ($79.1 \text{ kg N ha}^{-1}$, 31.1%, Fig. 5, Supplemental Information 9, Supplemental Information 10). LID showed significantly higher NH_3 volatilization compared to SLU (Supplemental Information 4).

All treatments showed a positive $\text{NH}_4\text{-N}$ balance (SLU: $73.6 \text{ kg N ha}^{-1}$, SLA: $66.9 \text{ kg N ha}^{-1}$, LID: $55.4 \text{ kg N ha}^{-1}$), indicating that the input of $\text{NH}_4\text{-N}$ was higher than the sum of the recovered $\text{NH}_4\text{-N}$ in the crop and soil and accounted losses (NH_3 emissions, NO_3^- leaching). The surplus of the balances can be attributed to pathways, that were not assessed, mainly denitrification and translocation to deeper soil layers, since only the topsoil was sampled.

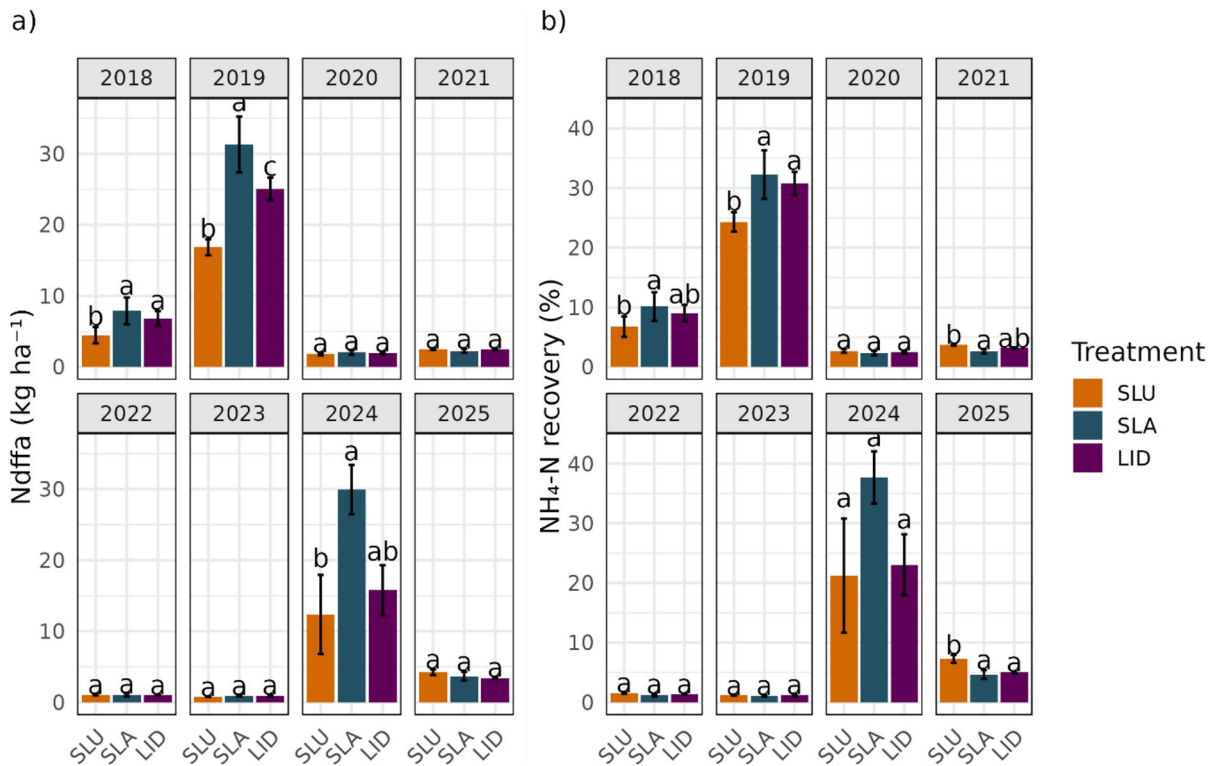


Fig. 3 Mean values ($\pm$ SD) of **a** nitrogen (N) derived from fertilizer ammonium N (Ndffa) in crops (kg N ha⁻¹) and **b** NH₄-N fertilizer recovery (%) from lysimeters under different fertilization treatments: untreated slurry (SLU), anaerobically digested slurry (SLA), and liquid digestate from non-agricultural biogas

plant (LID). The fertilizer treatments were labeled in 2018, 2019 and 2024. Different letters indicate significant differences between treatments within the year ($p < 0.05$, Tukey HSD). Maize was grown in 2018 and 2022, winter wheat in 2019 and 2023, winter barley in 2020, and 2025 and forage grass in 2021 and 2025

¹⁵N recovered in soil of the field experiment

In the field trial, only little NH₄-N was translocated to deeper soil layers (0.4–0.6 m, 0.6–1.0 m). In 2025 (residual year seven, Fig. 6) MP18 still captured most of the recovered ¹⁵N in the two uppermost soil layers (0.0–0.2 m, 0.2–0.4 m). In the layer 0.0–0.2 m of MP18 most ¹⁵N was recovered under SLU (4.9 kg $\pm$ 0.9 kg N ha⁻¹). Regardless of the treatment, less than 0.4 kg N ha⁻¹ was recovered in the deeper soil layers in MP18. Over the entire soil profile, the values ranged from 4.7 kg N ha⁻¹ under LID to 7.1 kg N ha⁻¹ under SLU. Similarly, MP19 showed the highest ¹⁵N recovery in the two uppermost soil layers (0.0–0.2 m, 0.2–0.4 m), and less than 0.4 kg N ha⁻¹ in deeper soil layers, with 7.0 kg N ha⁻¹ (LID) to 9.0 kg N ha⁻¹ (SLA) in the whole soil profile. The ¹⁵N recovered in the 0.0–0.2 m top layer in the first residual year of MP24, was 35.9 kg N ha⁻¹ in SLA

and 23.6 kg N ha⁻¹ under SLU, with less than 0.7 kg N ha⁻¹ translocated to deeper soil layers.

Discussion

Yields on lysimeter were representative for Swiss organic farming systems

Overall, fertilized crop yields were representative of Swiss organic farming systems, although yields varied considerably due to contrasting weather conditions. Crop yields varied considerably between years, reflecting the influence of weather extremes. The mixed-effects models confirmed a strong effect of year on dry matter yield. Maize performance was particularly affected by drought in 2018 resulting in reduced yield compared to 2022, when Maize was cultivated in a relatively normal year. Compared to

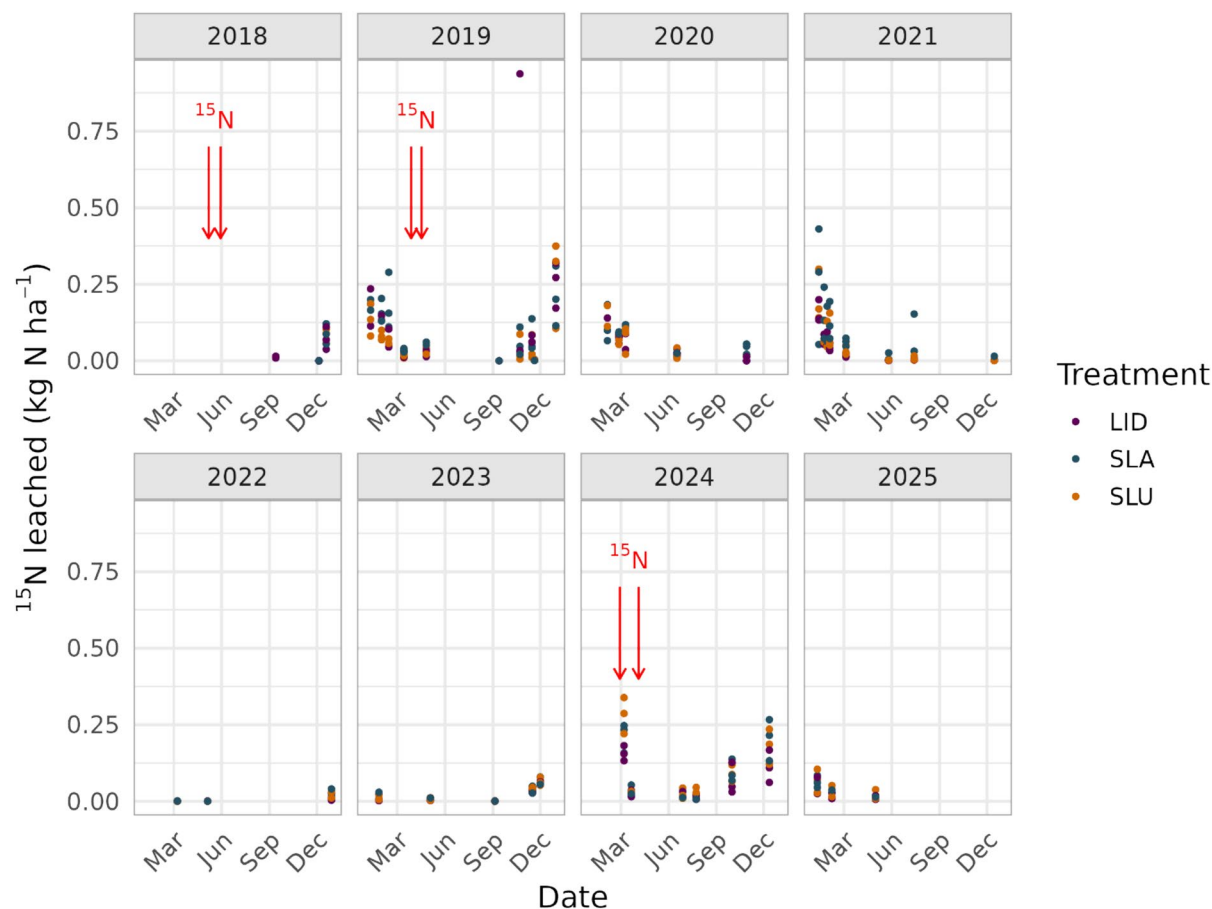


Fig. 4 Yearly timeseries of ^{15}N leached in kg N ha^{-1} from $\text{NH}_4\text{-N}$ in fertilizer in collected seepage water from lysimeters under different fertilization treatments: zero-N fertilization control (NON), untreated slurry (SLU), anaerobically digested

slurry (SLA), and liquid digestate from non-agricultural biogas plant (LID). Red arrows indicate ^{15}N labeling of the liquid organic fertilizers in 2018, 2019, and 2024

Table 5 Soil organic carbon (C) and nitrogen (N) concentration (g kg^{-1}), total N in soil (kg N ha^{-1}), N derived from fertilizer-ammonium ($\text{NH}_4\text{-N}$) in soil (Ndffa in soil, kg N ha^{-1}) and $\text{NH}_4\text{-N}$ fertilizer recovery (%) in topsoil (0–0.2 m) from lysim-

eters under different fertilization treatments: untreated slurry (SLU), anaerobically digested slurry (SLA) and liquid digestate from non-agricultural biogas plant (LID) eight years after start of lysimeter experiment

	Soil organic C (g kg^{-1})	Soil organic N (g kg^{-1})	Total N in soil (kg N ha^{-1})	Ndffa in soil (kg N ha^{-1})	$\text{NH}_4\text{-N}$ fertilizer recovery (%)
NON	10.2 ± 0.1 (b)	0.76 ± 0.02 (b)	2136 ± 61 (b)	–	–
SLU	14.0 ± 0.8 (a)	1.07 ± 0.09 (a)	3007 ± 243 (a)	44.9 ± 6.4 (a)	23.2 ± 3.3 (a)
SLA	13.7 ± 0.4 (a)	1.06 ± 0.09 (a)	2968 ± 24 (a)	46.3 ± 1.3 (a)	18.2 ± 0.5 (a)
LID	13.7 ± 1.3 (a)	1.05 ± 0.05 (a)	2936 ± 140 (a)	48.5 ± 5.8 (a)	21.5 ± 2.6 (a)

long-term yield data from an organic system trial in Switzerland (Knapp et al. 2023), maize yields from the lysimeters were significantly lower in 2018, but average in 2022 (Supplemental Information 5).

Winter wheat yields achieved on the lysimeters were average in 2019 and exceeded the long-term organic yield in 2023 (Knapp et al. 2023). Winter barley performed well in 2020, exceeding standard reference

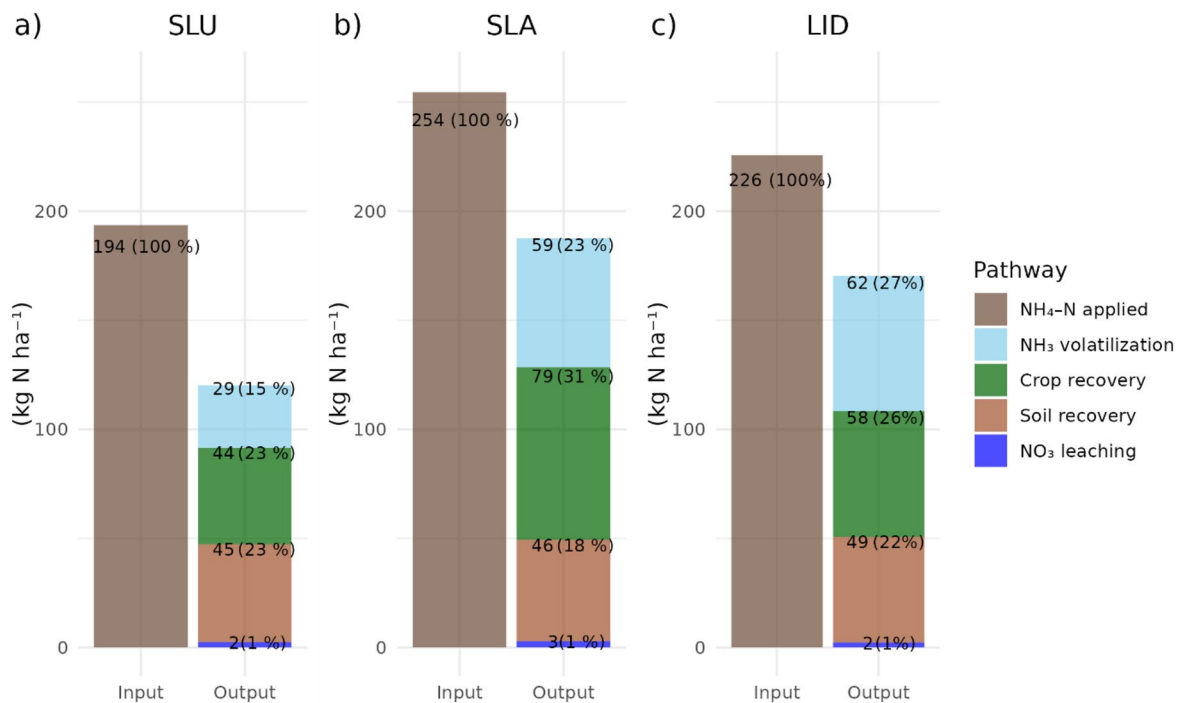


Fig. 5 NH₄-N balance from lysimeters under different fertilization treatments: **a** untreated slurry (SLU), **b** anaerobically digested slurry (SLA), and **c** liquid digestate from non-agricultural biogas plant (LID). NH₄-N applied is the NH₄-N input via fertilizers of the years, when ¹⁵N was applied (2018, 2019, 2024). NH₄-N output is structured into NH₃ emissions (modeled with the ALFAM2 model) at the applications (2018,

2019, 2024) indicated with light blue, ¹⁵N recovery by crops indicated with green, ¹⁵N recovery in the soil indicated with brown and ¹⁵N recovered in NO₃⁻ of seepage water indicated with dark blue. The values show the sum in kg N ha⁻¹ over the years of each pathway. The values in brackets show the NH₄-N losses respectively recovery (%) relative to the NH₄-N input

values for conventional farming systems of 6 t ha⁻¹ (Sinaj and Richner 2017).

Generally, crop yields from the lysimeters were higher than those in the mirrored field study with the same fertilizers and management (Efosa et al. 2025; Diener et al. 2026). Differences in yields between the lysimeter and the field can be explained by different pedoclimatic conditions and by the design of the lysimeter facility. The lysimeters had a surface area of 1 m² and were installed in a grassed area, which increased air circulation compared to field conditions. This likely reduced disease pressure and may have improved crop performance in wet years. For example, winter barley in 2024 likely benefited from the improved ventilation, as the season was characterized by high precipitation and severe disease pressure, in particular in organic farming systems (Diener et al. 2026). By contrast, the lysimeter setup also increased evapotranspiration, intensifying

drought stress in periods of low precipitation. This likely amplified drought effects and reduced maize yields in the dry growing season of 2018. Thus, the lysimeter design may have influenced crop performance in an opposing direction, depending whether disease pressure or water limitation was a stress factor. A study conducted in the same lysimeter facility showed consistently higher crop yields compared to average yields on Swiss farms throughout twelve years (Bernert et al. 2025), indicating overall beneficial effects of the lysimeter facility on crop performance.

N uptake and apparent NUE did not differ between digestates and untreated cattle slurry

Digestates did not consistently lead to higher N uptake or higher apparent NUE than untreated slurry, despite the higher proportion of plant available NH₄-N.

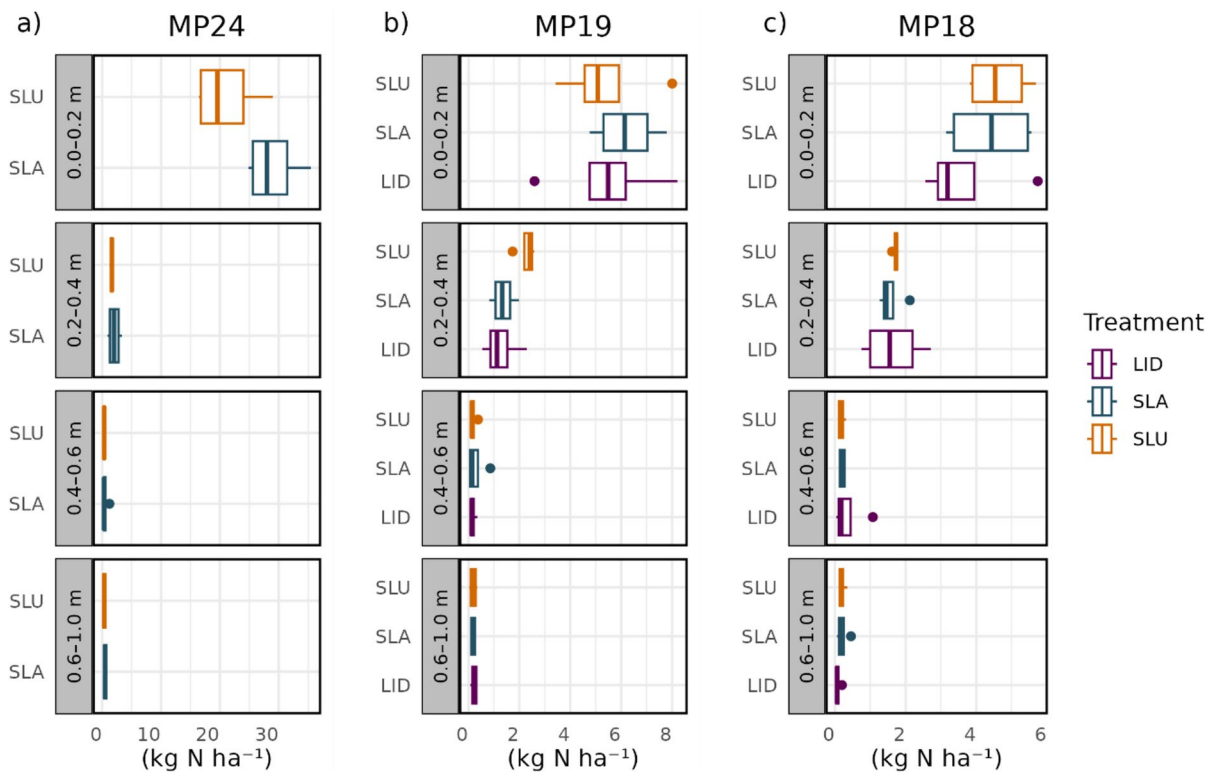


Fig. 6 Recovery of ^{15}N in kg N ha^{-1} of applied in $\text{NH}_4\text{-N}$ in the field along the soil profile (0.0–0.2 m, 0.2–0.4 m, 0.4–0.6 m, 0.6–1.0 m) of untreated slurry (SLU), anaerobically digested slurry (SLA), and liquid digestate from non-agricultural biogas plant (LID). **a** Microplot 24 (MP24) was labeled in 2024 with $(^{15}\text{NH}_4)_2\text{SO}_4$ (99 atom%) to 7 atom% excess, whereas **b** microplot 19 (MP19) was labeled in 2019 and **c**

microplot 2018 was labeled in 2018. In MP24 only SLA and SLU were labelled, due to the experimental set up. Boxes represent the interquartile range (25th–75th percentiles), the horizontal line indicates the median, whiskers extend to $1.5 \times$ the interquartile range, and points beyond the whiskers represent outliers

Although the differences were not statistically significant, anaerobically digested slurry resulted in higher NUE than untreated cattle slurry in winter cereals, whereas the liquid digestate from the non-agricultural biogas plant showed no improvement or decrease in NUE compared with untreated cattle slurry. Higher $\text{NH}_4\text{-N}$ proportion of digestates does not necessarily increase NUE, since higher pH often counterbalances the positive effects via losses such as NH_3 (Möller and Müller 2012; Pedersen and Hafner 2023). Modeled NH_3 emissions (Hafner et al. 2025) in this study were indeed higher for digestates (SLA: 23% of applied $\text{NH}_4\text{-N}$, LID: 27% of applied $\text{NH}_4\text{-N}$) than for untreated slurry (SLU: 15% of applied $\text{NH}_4\text{-N}$). Möller et al. (2008) reported positive effects of digestates on N uptake, only when they were immediately incorporated into the soil, which reduced NH_3 losses

and highlights the importance of low emission application techniques for digestate application. In our experiment, organic fertilizers were not incorporated, which is standard in Switzerland. Measured values in the field (Wallbach, Switzerland) showed higher but not significantly different NH_3 emissions for digestates (SLA: 42% of applied $\text{NH}_4\text{-N}$, LID: 43% of applied $\text{NH}_4\text{-N}$) compared to untreated slurry (SLU: 31% of applied $\text{NH}_4\text{-N}$; Efosa et al. 2025). These values obtained in the field are higher than the modeled emissions at the lysimeter facility, suggesting that real NH_3 emissions might be even slightly higher, despite differences in weather conditions. Digestates are typically characterized by a dry matter content between 4.5% and 6.6% (Möller and Müller 2012), but the liquid digestate from the non-agricultural biogas plant had a substantially higher dry matter content (10.4%)

due to the addition of co-substrates. The higher dry matter content may have impaired infiltration rate into the soil, thereby contributing to increased NH_3 emissions (Pedersen and Hafner 2023).

Organic fertilizers caused relatively low NO_3^- leaching over eight years

Annual NO_3^- leaching was between 12 kg N ha^{-1} for LID and 18 kg N ha^{-1} for SLA, which is relatively low compared to 38 kg N ha^{-1} for reference values in Swiss arable land calculated with the MODIFUS model (Hutchings et al. 2023). The annual NO_3^- losses measured in the lysimeters were slightly lower, than those reported from organic arable farming systems across Europe, where values ranged from 19 kg N ha^{-1} in Denmark (Hansen et al. 2000) to 24 kg N ha^{-1} in Germany (Biernat et al. 2020). A study by Bernert et al. (2025) performed on the same lysimeter facility showed mean annual leaching of even lower amounts (11 kg N ha^{-1}), when minerally fertilized according to Swiss fertilization recommendations (Sinaj and Richner 2017). Interestingly, annual NO_3^- leaching remained identical when fertilization was 130% of the recommended amount, because of higher crop N uptake (Bernert et al. 2025). This generally indicates good growth conditions and low NO_3^- leaching under the pedoclimatic conditions of the lysimeter facility.

Organic fertilization did not increase NO_3^- leaching compared with unfertilized crops at the applied N application rates. Mixed-effect models showed that interannual variation accounted for a larger proportion of the variability in NO_3^- leaching than differences among fertilizer treatments. The observed interannual variability may have been related to differences in annual precipitation patterns and crop characteristics across the rotation.

Fertilization amounts in this study were based on total N applied, whereas conventional farmers often fertilize based on effective N availability ($\text{NH}_4\text{-N}$ and mineralization of the previous year). In case of cattle slurries, this might result in doubled amounts of total N applied, since approximately 45% is immediately available to plants (Gutser et al. 2005; Epper et al. 2025). Especially, in intensive conventional farming systems, the applied N rates are higher than those in this study (Seufert et al. 2012; Krause et al. 2024). Fertilizer application rates exceeding crop N demand

exponentially increase NO_3^- leaching (Delin and Stenberg 2014; Di and Cameron 2002). Therefore, identifying the crop- and site-specific optimum of N application rates is crucial balancing crop yield and NO_3^- losses.

In this experiment, digestates had higher total N concentrations than untreated cattle slurry, because of the addition of co-substrates. Given that farmers frequently apply liquid organic fertilizers based on fixed application volume rather than on N concentrations, digestate applications might result in higher N inputs. In Switzerland, for example, application rates of $30 \text{ m}^3 \text{ ha}^{-1}$ per application are common (Sinaj and Richner 2017). Consequently, the application of these digestates without adjusting fertilizer application volumes may increase NO_3^- leaching (Di and Cameron 2002).

Bare soils are prone to NO_3^- leaching because the absence of an actively growing plant allows mineral N to accumulate in the soil and become susceptible to NO_3^- leaching (Tonitto et al. 2006; Thapa et al. 2018). At the fertilization level of this study, unfertilized crops exhibited NO_3^- leaching comparable to that of fertilized crops, indicating that the absence of N fertilizer input was offset by increased seepage water volumes resulting from reduced crop growth due to N deficiency. In contrast, fertilized crops efficiently acquired the applied N, likely preventing a substantial accumulation of mineral N in the soil while enhanced crop development increased water uptake and evapotranspiration. Nevertheless, both fertilized and unfertilized crops showed the highest NO_3^- leaching during winter, highlighting the importance of maintaining plant cover during this period to mitigate NO_3^- losses (Sørensen et al. 2023; Thapa et al. 2018).

NO_3^- leaching from digestates did not differ from untreated cattle slurry

The application of digestates showed contrasting effects on cumulative NO_3^- leaching over the eight-year period. Although neither digestate differed from untreated cattle slurry, the anaerobically digested slurry resulted in significantly higher cumulative NO_3^- leaching than the liquid digestate from the non-agricultural biogas plant. Inconsistent effects of digestate application on NO_3^- leaching are in line with other studies, showing either no (Zilio et al.

2023; Frick et al. 2024) or reducing (Sogn et al. 2018) effects compared to untreated cattle slurry. Different effects on NO_3^- leaching highlight the importance of co-substrates used in the biogas plant in affecting the fertilizer value of digestates (Möller and Müller 2012). In the present study, reduced NO_3^- losses of LID can be partially explained by the modeled NH_3 emissions at application (Hafner et al. 2019; Efosa et al. 2025). Given that 27% of the applied $\text{NH}_4\text{-N}$ in LID was already lost at application and did not reach the soil–plant system, NO_3^- leaching was decreased. When using digestates with a high dry matter content and high pH, a reduction of NO_3^- leaching may come at the expense of increased NH_3 volatilization (Pedersen and Hafner 2023). The low cumulative NO_3^- leaching and ^{15}N recovery in the seepage water indicate that NO_3^- leaching represents only a minor pathway of N losses from digestates compared with NH_3 emissions, further emphasizing the importance of good agricultural management to minimize NH_3 emissions.

Soil N pool was important for N dynamics of organic fertilizers

The soil organic N pool played an important role in N cycling in this study. Annual NO_3^- leaching without fertilization (16 kg N ha^{-1}) was similar to that with organic fertilization ($12\text{--}18 \text{ kg N ha}^{-1}$), indicating a substantial contribution of soil N mineralization to total NO_3^- leaching. Likewise, ^{15}N recovery in seepage water was low for all treatments (Fig. 4), emphasizing the small contribution of fertilizer $\text{NH}_4\text{-N}$ on immediate NO_3^- leaching. This agrees with Frick et al. (2022), who reported that short-term NO_3^- leaching from cattle slurry was minimal in the first 2.5 years after application. Even when both the organic N and $\text{NH}_4\text{-N}$ fractions of the fertilizers were ^{15}N labeled, less than 5% of leached NO_3^- originated from fertilizer, whereas the majority originated from mineralization of soil organic N (Frick et al. 2022).

Following organic fertilizer application, a part of $\text{NH}_4\text{-N}$ is immobilized within days into the microbial N pool and further incorporated into the soil organic N pool by the end of the application year (Frick et al. 2023; Epper et al. 2025). After eight years of this study, 18–23% of the applied $\text{NH}_4\text{-N}$ was recovered in the soil, highlighting the importance of the soil as a N reservoir. In the years of ^{15}N labeling in 2019 and

2024, crops recovered between 21% (SLU) and 38% (SLA) of applied $\text{NH}_4\text{-N}$, whereas only 2–5% was recovered in the subsequent year. This is consistent with previous ^{15}N studies that reported high fertilizer N recovery by crops in the year of application but less than 2% in the following year (Sebilo et al. 2013; Fuchs et al. 2023).

As only the $\text{NH}_4\text{-N}$ fraction of the fertilizers was ^{15}N labeled, the contribution of fertilizer organic N to NO_3^- leaching could not be quantified. However, because NO_3^- leaching did not differ between fertilized and unfertilized lysimeters, the contribution of organic N might have been small during the studied period. Repeated organic fertilizer applications build up the soil organic N pool, contributing to long-term effects on leaching after remineralization (Bergström and Kirchmann 1999), which has to be accounted in adjusted fertilizer management to prevent overfertilization. Using ^{15}N labeled mineral fertilizer, Sebilo et al. (2013) demonstrated that after nearly 30 years, still small amounts of leached NO_3^- originated from the fertilizer, highlighting the long-term effect of fertilization on NO_3^- leaching. Similarly, a lysimeter study in Denmark reported that 2% of NO_3^- leached from residual ^{15}N remaining in soil 25 years after application of ^{15}N labelled fertilizers (Sørensen et al. 2023). Consistent with these findings, 5–10% of applied $\text{NH}_4\text{-N}$ was still recovered in the soil six years after ^{15}N application in our field trial, demonstrating the persistence of fertilizer-derived N in the soil and the long-term importance of the soil N pool for N dynamics.

The field data also indicate that there was only a minor translocation of ^{15}N from the $\text{NH}_4\text{-N}$ fraction of the organic fertilizers to deeper soil layers. A similar observation was made by Frick et al. (2022), who recovered 4–10% of ^{15}N in the 0.3–0.6 m soil horizon, and neglectable amounts in the 0.6–0.9 m soil horizon 2.5 years after the application of slurry with labeled organic N and $\text{NH}_4\text{-N}$. With large liquid manure application (300 kg N ha^{-1} to winter wheat), translocation of N to deeper soil layers was shown to increase but was still very small ($<1.0 \text{ kg N ha}^{-1}$) under temperate continental climate (550 mm annual mean precipitation, 12.9°C annual mean temperature) (Ma et al. 2022).

NH₄-N balances differed by fertilizers

The NH₄-N balances revealed that the NH₄-N fate pathways (NH₃ volatilization, plant uptake, leaching, and soil recovery) differed between treatments: the untreated cattle slurry showed the highest recovery in soil, the anaerobically digested slurry showed the highest plant uptake, and the liquid digestate from the non-agricultural biogas plant showed the highest NH₃ losses. Untreated cattle slurry resulted in the highest amount of unaccounted NH₄-N (73.7 kg N ha⁻¹), whereas either digestate exhibited smaller amounts (SLA: 66.9 kg N ha⁻¹, LID: 55.4 kg N ha⁻¹).

The NH₄-N surplus can be attributed to denitrification of fertilizer-NH₄-N to N₂O and N₂ and translocation to deeper soil layers (<0.2 m), which neither process was technically feasible to assess in this study. Since the field data and other studies (Frick et al. 2022) showed that translocation to deeper soil layers was a minor effect, differences in NH₄-N balances might be explained by denitrification to N₂O and N₂ of applied fertilizer N. Kleineidam et al. (2025) reviewed a mean N₂ emission factor of 10.3% (min: 1.6%, max: 34.2%) of applied fertilizer N for organic fertilizers, which would result in 41.2 kg N ha⁻¹ total N₂ emissions in the present experiment (total N applied in 2018, 2019 and 2024 = 400 kg N ha⁻¹; 400 * 10.3 /100 = 41.2 kg N ha⁻¹). Indeed, the NH₄-N balances show a surplus between 55.4 kg N ha⁻¹ (LID) and 73.7 kg N ha⁻¹ (SLU). In a lysimeter study, applying ¹⁵N tracer in grassland, N₂ emission factors for cattle slurry were lower (3.5% of applied fertilizer N), possibly due to the high N uptake of the grassland (Dannenmann et al. 2025). Velthof et al. (2009) estimated overall N₂ losses of 44 kg N ha⁻¹ yr⁻¹ on agricultural land, indicating that the estimated N₂ losses in the present study are plausible. Denitrification is affected by labile C compounds triggering microbial activity (Surey et al. 2020; Kleineidam et al. 2025); therefore, digestates tend to have lower N₂O and N₂ emissions than untreated slurry. Indeed, N₂O emissions of the mirrored experiment in the field were lower for digestates (Agostini et. al. unpublished), supporting the finding that denitrification losses in the lysimeters might be higher for untreated slurry compared to values for digestates, due to more available C in the slurry (Weier et al. 1993). Overall, the NH₄-N balances showed that regardless of the fertilizer treatments, only little N was lost through NO₃⁻ leaching,

whereas NH₃ volatilization and denitrification can significantly affect N fertilizer balances.

Conclusion

Organic fertilization increased plant N uptake and dry matter yield without increasing NO₃⁻ leaching compared with unfertilized crops over the eight-year period. However, N application rates in intensive systems are often higher than those in this study, which may lead to increased risk of NO₃⁻ leaching. The investigated digestates showed no distinct advantage over untreated cattle slurry in reducing NO₃⁻ leaching or improving apparent NUE across all crops. Organic fertilizer application had only a minor effect on NO₃⁻ leaching during the year of application, indicating that NO₃⁻ leaching was primarily controlled by the soil organic N pool, which retained approximately 20% of the applied NH₄-N. By contrast, digestates caused high modeled NH₃ losses at application, potentially hampering apparent NUE and causing pollution swapping between volatilized NH₃ and leached NO₃⁻. Cumulative NO₃⁻ leaching and recovered ¹⁵N in the seepage waater were small, revealing that NO₃⁻ leaching was a relatively minor fertilizer N loss pathway compared to NH₃ volatilization and losse through denitrification.

The findings highlight the importance of integrating the long-term remineralization of fertilizer N retained in the soil into crop- and site-specific fertilization strategies, to optimize productivity while minimizing N losses in form of NO₃⁻ leaching. Organic fertilizer application should also be accompanied by management strategies that maintain permanent soil cover, particularly during winter when the risk of NO₃⁻ leaching is highest in temperate climates. Future research should focus on the long-term (>20 years) application of organic fertilizers and associated N mineralization dynamics to improve predictions of NO₃⁻ leaching supporting sustainable fertilizer management.

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Data availability The data are available under the following link in Zenodo: <https://doi.org/10.5281/zenodo.21819861>.

Declarations

Conflict of interest The authors declare no competing interests.

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